

Assignment 1 – Individual assignment

Part A – SE Interview Preparation – 55 Points, 4 points each unless specified otherwise

For the interview questions especially, I recommend using it as an opportunity to actually practice by trying to speak/dictate/write an answer from memory, then look at slides and the readings to improve your answer before submitting.

1. What are the phases of the software development process? Briefly explain them.

Requirements Gathering: Finding out what the software needs to do.

Design: Planning how the software will work.

Implementation: Writing the code.

Testing: Checking if the software works correctly.

Deployment: Releasing the software for use.

Maintenance: Fixing problems and updating the software.

2. What is the difference between verification and validation? Which software development stages are they part of?

Verification involves ensuring the software is built correctly, and it takes place during the design and implementation stages. Validation, on the other hand, ensures the software meets the user's needs and occurs during the testing stage.

3. What are some benefits and limitations of the agile development process? What are some practices to address them?

Agile is great because it's flexible and lets teams make quick changes, keeping customers involved throughout the process. However, predicting timelines can be tough, and good communication is essential. To handle these challenges, teams should have regular meetings, communicate clearly, and use tools to track their progress.

4. Describe the different roles on a software development project.

Project Manager: Oversees the project.

Designers: Create the user interface.

Developers: Write the code.

Testers: Check for bugs and issues. (technically this is everyone)

5. What distinguishes a great software engineer from a good one? Do you feel you have those qualities?

A great software engineer is distinguished from a good one by their ability to solve problems effectively, communicate well, and continuously learn, whereas good engineers perform their tasks competently but may not go above and beyond. I believe I possess qualities such as problem-solving and teamwork that align with being a great engineer.

6. How do you prioritize which requirements to implement first?

Focus on the most important features that add the most value or are needed urgently.

7. How do you think about software development? What are the key challenges and how do you address those challenges when working?

Software development is about solving problems with code. Common challenges include unclear requirements and tight deadlines. To handle these challenges, it's important to communicate clearly, break tasks into smaller parts, and stay organized. By doing these things, developers can work more efficiently and effectively.

8. In your opinion, what are the principles of good software engineering? What are some basic principles everyone should follow?

Some basic principles that everyone should follow include; writing clean code, testing often, and keeping the end-users needs in mind.

9. How do you respond when you disagree with a team member?

When I disagree with a team member, I normally try to listen to what they are saying and try and find a way to meet their needs while also meeting my own.

10. What are some tensions and trust issues that can arise between software developers and business people / customers? What are ways to address those tensions?

Tensions and trust issues between software developers and business people/customers often arise from misunderstandings about requirements or deadlines. It's important to have regular communication and set clear expectations about the project's progress. This helps build trust and ensures everyone is on the same page.

11. (5 points) Can you answer any of these interview questions well yet? You may either answer each one, or alternatively describe in 1-2 sentences how you can remember and have good answers to each by the end of this course.

1. Have you ever had a problem with another team member that you were working with? If so, how did you handle it?

Yes, there was an occurrence where me and a team member ran into an issue where we did not agree on an idea for the outline of our homework. We were able to discuss and hear each other's thoughts and come up with a solution incorporating both ideas.

2. Describe a time when you had to overcome a significant obstacle on a software engineering project.

One time that I was having issues with a software engineering project was when I was trying to assist someone to troubleshoot an issue, but there was not sufficient documentation. This made it very difficult to assist with the troubleshooting.

3. Why were you successful in your last software engineering project? Give a supporting example.

I would say that it was successful because me and my friend that I did the project with both had pretty decent comments so when there was anything that we had a problem with anything, we were able to tackle the problem quickly.

5. (10 Points) Are the following requirements good requirements? If not, why? Correct Them.

- A. The system shall be secure enough to send messages between users and be available at all times.**

The system shall use encryption to secure messages and maintain a 99.9% uptime.

- B. The system shall recognize the user through face recognition and allow them to add friends when joined.**

The system shall authenticate users with facial recognition and enable them to add friends upon a successful login.

- C. The system shall begin a file download within 1 second of a request 99% of the time.**

This one is correct.

Part B – User Stories and Functional and Non-Functional Requirements – 45 Points

You are the lead software developer at Les-Animaux company that develops mobile applications.

Currently, they are developing a dog sitting application called, DogCare.

Les-Animaux's main high-level goals are to make profit through advertising, be fast in the market, minimize its development costs, protect the privacy of the individuals and ensure security of the application. The app has two versions: free and paid.

In DogCare application, there are two types of users: (1) Dog Sitters: the users that would like to work as dog sitters; and (2) Dog Owners: the users that need someone to take care of their dog(s) when they are away for holidays or just during the day, when they are at work.

Both users first need to register, and enter their personal information such as name, age, gender, location/address, hobbies, and why they are interested in dogs.

The dog owners also add information about their dogs: number of them, their breeds, the types of food/treats they eat/like, ages, any health restrictions, etc. The dog sitters need to also enter their availability, price per hour, and other restrictions. Both have options to upload their photos or the photos of their pets.

The app works similar to dating apps by creating a match between a dog owner and a dog sitter. Both can decide if they like each other or not (i.e. swipe left if they do not like each other). If one of them likes the other, they can send a friend request. The other user needs to approve or reject the request before they make connections.

The app tracks the location of the users. The users have an option to limit the tracking to *always, only once, while using the application* or *never*. If a user selects never, the application is not usable, though. The dog owners can also decide to ask to track the location of their dogs when they are with the dog sitter, at all times or at predefined intervals. This is an add-on feature which costs extra. On the other hand, the dog sitters can reject to be tracked all the time but then they may not get enough customers.

The dog sitters can use the free app if they do not want to make money from dog sitting. If the dog sitters want to make money, then they must pay a monthly or yearly fee for the application. Similarly, the dog owners can use the free application if they only need dog sitters twice per month. For more than that, they need to pay monthly or yearly fees. DogCare also collects SSN and banking information from dog sitters for tax purposes if they use the paid application. The dog owners also should provide their banking information if they use the paid application.

To increase profits, DogCare might share data with certain third parties for advertising purposes.

Les-Animaux needs to make sure to protect the privacy and security of the collected data. In addition, since the company operates both in California and EU, the California Consumer Privacy Act (CCPA) and the General Data Protection Regulation (GDPR) applies to the application.

(In answering this question, you may do this [practice assignment](#) separately briefly, and/or look at the [solutions](#))

1. **(15 Points)** Write 10 user stories for different features of the DogCare application. **Make sure to number your user stories.**

User Stories:

1. As a dog owner, I want to create a profile with my personal information and my dog's details so potential dog sitters can get to know us better.
2. As a dog owner, I want to search for dog sitters based on location and availability to find someone nearby to take care of my dog.
3. As a dog owner, I want to be able to track the location of my dog while it is with the dog sitter for safety.
4. As a dog owner, I want to leave a review and rating for a dog sitter after a session to share my feedback with other owners.
5. As a dog sitter, I want to create a profile with my availability, pricing, and restrictions so dog owners can see when I am available and how much I charge.
6. As a dog sitter, I want to receive notifications when a dog owner sends me a friend request so I can decide whether to accept or reject it.
7. As a dog sitter, I want to update my profile with new availability and rates to keep information current.
8. As a dog sitter, I want to have a calendar view of my bookings to manage my schedule efficiently.
9. As a dog sitter, I want to filter dog sitters by various criteria, such as location, price, and availability, to find the best match for my needs.
10. As a dog sitter, I want the ability to accept or decline booking requests from dog owners to manage my workload.

2. **(30 Points)** Write 10 functional requirements and 10 non-functional requirements for DogCare. Make sure to have at least 3 privacy and security requirements. **Make sure to number your requirements.**

Functional Requirements:

1. The system shall allow users to register and input their personal information.
2. The system shall allow dog owners to enter information about their dogs.
3. The system shall allow dog owners to schedule appointments with dog sitters.
4. The system shall enable dog sitters to enter their availability and pricing.
5. The system shall allow users to upload photos to their profiles.
6. The system shall enable users to swipe left or right on profiles to indicate interest.
7. The system shall send a friend request when one user likes another.
8. The system shall let users review the dog sitters after each session.
9. The system shall provide different options for location tracking.
10. The system shall allow dog owners to track their dogs' locations as an add-on feature.

Non-functional Requirements:

1. The system shall respond to user actions within 2 seconds to ensure a smooth user experience.
2. The system shall support high traffic volumes, handling up to 10,000 concurrent users.
3. The system shall have a user-friendly interface that is easy to navigate.
4. The system shall be compatible with both iOS and Android operating systems.
5. The system shall provide multi-language support, including English, Spanish, and French.
6. The system shall allow for easy integration with third-party payment gateways.
7. The system shall regularly back up user data to prevent data loss.
8. The system shall provide detailed error messages and troubleshooting steps for common issues.
9. The system shall ensure that updates and new features do not disrupt ongoing services.
10. The system shall require strong password policies for user accounts.